PHAS0050 : Climate & Energy
Module 2 : Nuclear Power

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1 Introduction

By the outbreak of the Second World War in 1939 it was known that thermal (KE ≈ 0.025 eV) and fast (KE $0.5 - 10$ MeV) neutrons would fission the ${}^{235}_{92}$U nucleus with the emission of 2.5 fast neutrons on average, the fission probability being particularly large for thermal neutrons. The fast neutrons released are also accompanied by 0.6% delayed neutrons. It was widely recognized that a chain reaction based on the fast neutrons could lead to a nuclear explosion with an energy release some 50 million times that of a chemical explosion. But it was two refugee academics at Birmingham University, Otto Frisch and Rudolf Peierls who realized in 1940 that an atomic bomb was possible using a few kilograms of ${}^{235}_{92}$U. Allied scientists and a huge industrial effort in the US produced the Hiroshima $\left({}^{235}_{92}\text{U}\right)$ and the Nagasaki $\left({}^{239}_{94}\text{Pu}\right)$ bombs which brought WW2 to a terrifying and tragic end in 1945. No nuclear weapon has been used offensively since that date.

Controlled energy release, by thermalising the fast neutrons and using the delayed neutrons for control (which ultimately led to thermal nuclear reactors) was used in reactor piles to produce ${}^{239}_{94}$Pu during WW2. The development of thermal (neutron) power reactors started after the end of hostilities and it was in the 1950's that reactors using fast neutrons, the so called Fast Beeder Reactors (FBR) were also developed.

In the US during the 1960s and 1970's a large scale commercial civilian nuclear power industry based on light water moderated (see later) thermal reactors developed for submarine propulsion was created. The CO_2 gas cooled, graphite moderated reactor developed from the graphite piles used to produce ${}^{239}_{94}$Pu was the basis for the first generation of MAGNOX power stations in the UK.

Initial confidence in the nuclear industry was eroded due to fears regarding safety, nuclear waste disposal and the dangers of nuclear weapon proliferation. There were costly engineering overuns and the nuclear power reactor accidents at Three Mile Island (1979), Chernobyl (1986) leading to a large release of radioactive material and Fukushima (2011). (Note that Windscale (1957) accident was in an air cooled pile producing weapons grade plutonium, not a power reactor.) By the 1990's energy utilities in the UK and US were becoming deregulated, restructured and where necessary, privatised which brought the market fully to bear on energy engineering. The days of engineering led design and cost plus contracts were gone for ever. The nadir of the nuclear industry in the US probably occurred in 1992 when the completed, fuelled and tested Shoreham nuclear plant on Long Island, New York State, proposed in 1965 was cancelled after an expenditure of 6 billion dollars without a kWh of electricity ever being sold. Inevitably the lack of new reactor orders has led to a decline in engineering expertise in the US and the UK whereas it was the breakdown of the command economy in the former Soviet Union that damaged the Russian capability. Even in Canada the dearth in orders has led to the state backed AECL company behind the successful CANDU generation of reactors being sold for a derisory sum to a private company. No such loss of expertise occurred in France where 80% of the electricity supply is nuclear generated although the lack of new build has probably been partly responsible for construction problems of the new generation of EPR reactors. India has a thriving largely home grown nuclear industry based in part on the CANDU reactor design and China is actively developing a nuclear power industry of its own. These countries are well placed to lead and to profit from the "Nuclear Renaissance" made possible by the concerns surrounding climate change and carbon emissions. It is difficult to see how the privatised companies in the developed world can make the long term investments required to develop the nuclear program without at the very least a floor price guarantee.

2 Fission and its energy release

Fission was first discovered when a beam of thermal neutrons (with kinetic energy ≈ 0.025 eV) interacted with natural uranium containing 99.3% ${}^{238}_{92}$U and $0.7\%{}^{235}_{92}$U where the${}^{235}_{92}$U fissioned into a range of lighter nuclei, the most abundant pair being isotopes of krypton $\left({}^{92}_{36}\text{Kr}\right)$ and barium $\left({}^{141}_{56}\text{Ba}\right)$ plus on average 2.5 prompt neutrons.

The ${}^{235}_{92}$U nucleus contains 92 protons (the atomic number, $Z = 92$), 235 nucleons (the mass number,

$A = 235$) and therefore 143 neutrons (the neutron number, $N = 143$). To a good approximation nuclear radii are given by the formula $R = 1.5 \times A^{\frac{1}{3}}$ fm where 1 fm ("Fermi") $= 1 \times 10^{-13}$ cm $= 1 \times 10^{-15}$ m. Thus the geometric cross sectional area presented by the uranium nucleus is $\pi\ (1.5 \times 6.17)^2\ \times 10^{-26}\ \mathrm{cm}^2$ $= 2.7 \times 10^{-24}\ \mathrm{cm}^2 = 2.7$ barns.

As a function of the number of nucleons (i.e. the mass number A) the binding energy of a nucleon bound in a nucleus increases up to the most stable nucleus, the isotope of iron ${}^{26}_{56}$Fe. Thereafter the nucleon binding energy decreases so that when ${}^{235}_{92}$U$+n$ (i.e. ${}^{236}_{92}$U) fissons into the more tightly bound ${}^{92}_{36}$Kr and ${}^{141}_{56}$Ba nuclei then there is a release of energy. The size of the energy release can be calculated either by noting that for uranium the binding energy per nucleon is 7.5 MeV whereas in the krypton barium region it is 8.4 MeV, or from the known nuclear masses of the uranium and fission daughter nuclei. Both estimates are close to 200 MeV.

More insight into fission can be obtained by considering that immediately after fission the krypton and barium nuclei are just touching. At this separation there is a repulsive electrostatic force between the 36 krypton protons and the 56 protons in the barium nucleus. As these nuclei fly apart the kinetic energy released corresponds to the potential energy of one of the nuclei in the electric potential of the other. At fission the centres of the nuclei are $(1.5\times 92^{\frac{1}{3}}+1.5\times 141^{\frac{1}{3}})$ fm apart, and so the energy released $\mathrm{K.E.}_{\text{fiss. frag.}}$ is given by :

$$\mathrm{KE}_{\text{fiss. frag.}} = \frac{1}{4\pi\epsilon_0}\frac{36e \times 56e}{1.5 \times 9.7} \approx 200\ \ \mathrm{MeV}\ \ ,$$

where e is the magnitude of the electron (and proton) charge, $\epsilon_0 = 8.85 \times 10^{-12}\ \mathrm{F\,m^{-1}}$ is the permittivity constant and in more convenient units $e^2/4\pi\epsilon_0 = 1.44\ \mathrm{MeV\,fm}$. It has been assumed that the charge distribution of the fission fragments is spherically symmetric. This is a crude estimate but serves to show that most of the energy released in fission is electrostatic in origin.

Fission fragment nuclei are neutron rich and $\beta-$decay into a daughter nucleus, an electron and an anti-neutrino or more rarely into a daughter nucleus plus neutron, with half-lives of the order of tens of seconds. As an example ${}^{87}_{35}$Br decays with a 55.6 second half-life as follows :

$${}^{87}_{35}\mathrm{Br} \rightarrow {}^{86}_{36}\mathrm{Kr} + n + e^- + \bar{\nu}_e \quad (2\%) \text{ and}$$

$${}^{87}_{35}\mathrm{Br} \rightarrow {}^{87}_{36}\mathrm{Kr} + e^- + \bar{\nu}_e \quad (98\%)\ .$$

These decay chains continue until stable daughter nuclei are reached and are the sources of the delayed neutrons so vital in the control of thermal and fast reactors.

Table 1 summarises the distribution of energy released from the neutron induced fission of ${}^{235}_{92}$U. From now on we assume that the neutron induced fission of ${}^{235}_{92}$U releases 195 MeV of energy, an average 2.5 prompt neutrons of mean energy 2 MeV and 0.7% delayed neutrons. Note that some of the neutrons will be captured on nuclei without causing further fission, and including this small additional energy contribution may increase the 195 MeV to approximately 200 MeV per fission. Either number is acceptable for performing fission energy yield calculations.

Fission Fragments	168 MeV
Prompt Neutrons	5 MeV
Prompt γ emission	7 MeV
β decay electrons	8 MeV
β decay neutrinos	12 MeV
Delayed γ emission	7 MeV
Total Visible Energy	195 MeV

Table 1: Energy release from neutron induced fission in ${}^{235}_{92}$U.

How much energy would be released by fissioning 1 kg of ${}^{235}_{92}\mathrm{U}$? A mole of ${}^{235}_{92}\mathrm{U}$ contains 235 g and contains Avogadro's number of atoms (i.e. $6.02 \times 10^{23}\ \mathrm{mol}^{-1}$), so the energy release is :

$$\frac{1}{0.235} \times 6.02 \times 10^{23} \times 195 \times 10^{6} \times 1.6 \times 10^{-19}\ \mathrm{J} = 8.0 \times 10^{13}\ \mathrm{J}.$$

Now suppose the this energy is produced in a day which contains $60 \times 60 \times 24$ seconds ie 8.64×10^{4} s then the thermal power is :

$$\frac{8.0 \times 10^{13}}{8.64 \times 10^{4}} = 0.926 \times 10^{9}\ \mathrm{W} \approx 1\mathrm{GW_t}.$$

For 'ball park' estimates we see that to produce a thermal energy output of $1\ \mathrm{GW_t d}$, i.e. a power output of $1\ \mathrm{GW_t}$ for a day, about 1 kg of a fissionable (see below) nucleus must be fissioned, the more exact number being 1.08 kg per day. This is not (quite) the amount of ${}^{235}_{92}\mathrm{U}$ consumed in a reactor, for reasons to be discussed later.

3 Cross sections, neutron flux and interaction length

The *cross section* σ_i for a particular process to occur is an imaginary area surrounding a target particle (in our case a nucleus) though which an incident particle (in our case a neutron) must pass so that the process takes place. These quantities are measured by experiment and predicted by theoreticians if a theoretical understanding exists.

Imagine a beam of neutrons of number density $n_b\ (\mathrm{m}^{-3})$, speed $v\ (\mathrm{m\,s}^{-1})$ and cross section $A\ (\mathrm{m}^2)$ incident on an infinite slab of thickness $dx\ (\mathrm{m})$ containing the target nuclei whose number density is $n_t\ (\mathrm{m}^{-3})$. Suppose each target nucleus is surrounded by the area σ_i, then the probability that one of the incident neutrons in the beam interacts with one nucleus in the target is σ_i/A. But in the target $n_t \times A \times dx$ nuclei are 'illuminated' by the beam, and in one second $n_b \times v \times A$ neutrons pass through the target so that the number of interactions d^2N_i in a time interval dt and target thickness dx (or volume element dV) is given by :

$$d^2N_i = n_b v A dt \times \frac{\sigma_i}{A} \times n_t A dx = n_b v \sigma_i n_t dV dt \text{ , or } \quad \frac{d^2N_i}{dtdV} = n_b v \sigma_i n_t = R_i \ ,$$

where dV is the target volume element illuminated by the beam. In words, the number of interactions per second (i.e. the rate) per unit volume R_i is equal to $n_b v \sigma_i n_t = \Phi \sigma_i n_t$ where $\Phi = n_b v\ (\mathrm{m}^{-2}\mathrm{s}^{-1})$ is called the *neutron flux* and is typically $10^{13} - 10^{14}\ \mathrm{cm}^{-2}\mathrm{s}^{-1}$ in a thermal reactor. Finally, it is sometimes useful to note that $\int n_t dV = N_t$ where N_t is the total number of target nuclei illuminated, and therefore providing Φ is uniform throughout we obtain :

$$\frac{dN_i}{dt} = \Phi \sigma_i N_t \ .$$

For thermal neutrons there are three neutron-nucleus cross sections that are important, namely the *scattering* cross section σ_s where the neutron scatters off the nucleus, the *capture* cross section σ_c where the neutron is captured by the nucleus and the *fission* cross section σ_f all of which add up to the total cross section σ_t. These cross sections are strongly energy and nucleus dependent as we shall see in Tables 2 and 3 below.

As we have seen above the uranium nucleus has a geometric cross section of $269 \times 10^{-26}\ \mathrm{cm}^2$ or $2.7 \times 10^{-24}\ \mathrm{cm}^2$ or 2.7 barns where the barn unit (1 barn $= 10^{-24}\ \mathrm{cm}^2$) was so named when a particular nuclear reaction was so large that the effective reaction cross section 'was as big as a barn door'. Scattering, capture and fission cross sections, which can be much smaller or indeed much larger than the geometric cross section are measured in barns.

Now we ask a slightly different question. Suppose a neutron travels a distance x through a material containing n_t nuclei per unit volume and each is imagined to be surrounded by a cross sectional area σ_i

through which the incident neutron has to pass, for a particular collision outcome to happen. How many such collisions happen on the neutron's trajectory which changes direction at each collision and is of total length x ? The cross section σ_i is imagined to surround the incident neutron (i.e. the cross sectional area is transferred from the target nuclei to the incident neutron), and collisions will occur with any nucleus (now point like) whose centre lies within the volume $\sigma_i x$ swept out by the cross section of the incident neutron. The number of collisions is $\sigma_i x n_t$ and the mean distance between these collisions is $x/\sigma_i x n_t = 1/\sigma_i n_t$. Thus any cross section can be associated with a *mean interaction length* λ_i via the relation $\lambda_i = 1/n_t\sigma_i$.

4 Fissile, fissionable and fertile nuclei

Fissile nuclei are those stable or long lived nuclei that are fissioned by thermal neutrons. There are three such nuclei that we are interested in, namely ${}^{233}_{92}\mathrm{U}$, ${}^{235}_{92}\mathrm{U}$ and ${}^{239}_{94}\mathrm{Pu}$.

Fissionable nuclei are those nuclei that can only be fissioned by fast neutrons. All the actinides including thorium ${}^{232}_{90}\mathrm{Th}$ and ${}^{238}_{92}\mathrm{U}$ are fissionable by fast neutrons having energies greater than about $1\ \mathrm{MeV}$.

Fertile nuclei are those nuclei that can be converted into fissile nuclei by thermal and/or fast neutrons. Examination of Table 2 shows that ${}^{238}_{92}\mathrm{U}$ has a large thermal and fast neutron cross section for neutron capture. After such a capture the following reaction and decays take place :

$$n + {}^{238}_{92}\mathrm{U} \rightarrow {}^{239}_{92}\mathrm{U}(23\mathrm{m}) \xrightarrow{\beta} {}^{239}_{93}\mathrm{Np}(2.36\mathrm{d}) \xrightarrow{\beta} {}^{239}_{94}\mathrm{Pu}(2.4\times 10^4\mathrm{yr})$$

Thorium ${}^{232}_{90}\mathrm{Th}$ which is not fissile can be converted by thermal and fast neutrons to ${}^{233}_{92}\mathrm{U}$ which is a fissile nucleus :

$$n + {}^{232}_{90}\mathrm{Th} \rightarrow {}^{233}_{90}\mathrm{Th}(22\mathrm{m}) \xrightarrow{\beta} {}^{233}_{91}\mathrm{Pa}(21\mathrm{d}) \xrightarrow{\beta} {}^{233}_{92}\mathrm{U}(1.6\times 10^5\mathrm{yr})$$

The properties of these nuclei are shown in Table 2 for thermal neutrons and in Table 3 for fast neutrons, where the properties are averaged over the appropriate neutron spectra, together with the fission cross section σ_f, the capture cross section σ_c and the scattering cross section σ_s again averaged over the neutron energy spectra. For the fast neutrons the inelastic cross section σ_{in} is added (see below). Note also that fast neutrons encounter large resonant cross sections which in fertile nuclei also lead to the production of fissile nuclei as the neutrons slow down.

	${}^{233}_{92}\mathrm{U}$	${}^{235}_{92}\mathrm{U}$	${}^{239}_{94}\mathrm{Pu}$	${}^{238}_{92}\mathrm{U}$	${}^{232}_{90}\mathrm{Th}$	$\mathrm{U_{NAT}}$
σ_f (barns)	469	507	746	-	-	3.56
σ_c (barns)	41	87	285	2.4	6.5	3.04
σ_s (barns)		10	9.6	9.4	13.7	9.4
ν	2.5	2.4	2.9	-	-	-
η	2.3	2.05	2.1	-	-	1.3
% Delayed neutrons	0.26	0.67	0.21	-	-	0.67

Table 2: *Thermal neutron* spectrum averaged cross sections and related parameters

In Table 2, ν is the average number of fast neutrons emited by fission and η is the number of fast neutrons emmited *per* neutron absorbed. The absorption of a slow neutron can lead either to fission or capture, so for a fissile species η and ν are related by :

$$\eta = \nu\frac{\sigma_f}{\sigma_f+\sigma_c} \text{ and for natural uranium } (0.007\%\ {}^{235}_{92}\mathrm{U})\ \eta_{\mathrm{NAT}} = \nu_5\frac{0.007\sigma_{5f}}{0.007(\sigma_{5f}+\sigma_{5c})+0.993\sigma_{8c}} = 1.3\ .$$

	$^{233}_{92}$U	$^{235}_{92}$U	$^{239}_{94}$Pu	$^{238}_{92}$U	$^{232}_{90}$Th	$\mathrm{U_{NAT}}$
σ_f (barns)	1.9	1.20	1.80	0.31	0.08	0.32
σ_c (barns)	0.07	0.09	0.05	0.07	0.09	0.07
σ_s (barns)	4.4	4.6	4.4	4.8	4.7	4.8
σ_{in} (barns)	1.2	1.8	1.5	2.6	2.9	2.6
ν	2.6	2.5	3.0	-	-	-
η	2.4	2.35	2.90	0.07	-	2.6
% Delayed neutrons	0.26	0.64	0.21	1.64	-	1.64

Table 3: *Fast neutron* spectrum averaged cross sections and related parameters

A similar weighting formula would give η for enriched fuel. Both ν and η are properties of the nuclear fuel alone, be it pure fissile material and natural or enriched uranium.

For the fast neutron table the inelastic cross section σ_{in} describes the process where fast neutrons inelastically scatter, exciting the target nucleus into a state which then decays back to the ground state with the emission of a gamma ray. This does not lead to a loss of neutrons, rather it leads to neutron energy loss.

For a thermal energy output of $1\ \mathrm{GW_t d}$ we learned above that $1.08\ \mathrm{kg}$ of a fissile material must be fissioned. The amount of fissile material *consumed* to produce $1\ \mathrm{GW_t d}$ is $(\sigma_f + \sigma_c)/\sigma_f$ times this amount, i.e. approximately $1.17\ \mathrm{kg}$ for $^{233}_{92}$U, $1.26\ \mathrm{kg}$ for $^{235}_{92}$U and $1.49\ \mathrm{kg}$ for $^{239}_{94}$Pu. But for many 'ball park' calculations it is sufficient to remember that to generate $1\ \mathrm{GW_t}$ thermal power requires the consumption of about $1\ \mathrm{kg}$ of fissile material per day.

Note also that as fast neutrons slow down in fertile nuclear materials large resonance capture processes (see previous page) are encountered which remove the neutrons before they thermalise, but create fissile nuclei.

5 Criticality, Control and Stability

A reactor must consist of power generating fissile material, possibly fertile material, other materials for structure and possibly neutron moderation (see below) and a gas, liquid or molten metal (even molten salt) to cool the reactor and to transfer the generated heat.

For power generation the first requirement is criticality, i.e. the neutron chain reaction be self sustaining, and the second is that control is achieved. *Criticality* and *control* are achieved by ensuring that the neutron multiplication factor k, which is the ratio of the number of neutrons producing fission to the number producing fission in the previous generation, be close to, but below unity such that criticality at $k = 1$ is achieved by the *delayed neutrons*. During a reactor's operation the criticality conditions will change so that *stability* has to be ensured by designing the *reactivity* $\rho = \frac{k-1}{k}$ to have an overall negative temperature coefficient (i.e. if the reactor temperature were to rise then the reactivity would decrease) and by the use of neutron absorbing *control rods* and movement of *neutron reflectors* on the reactor periphery. In water cooled reactors neutron absorbing liquid such as boric acid can be added to control the criticality. In the event of *emergency conditions* the reactor core could be filled with neutron absorbing material and it is essential to cool the fuel even after fissioning has ceased since roughly 6.5% of the reactor power is produced immediately after shutdown by the radioactive decay of the fission fragments in the fuel.

6 The Neutron Multiplication Factor k_∞ and k

Note carefully that this and the next section on moderators is based on physical insight; real reactor design must be based on much more quantitative analysis than this.

The neutron multiplication factor k_∞ applies to a reactor of infinite size and is the product of four factors (hence the designation **four-factor formula**) :

$$k_\infty = \eta \times \epsilon \times p \times f \ ,$$

where we shall see that two of the factors f and p depend on reactor design. The actual neutron multiplication factor k is related to k_∞ by $k = k_\infty(1 - l_f)(1 - l_t)$. Each term in the formula has the following meaning :

- η is as described above and is the average number of fast neutrons emitted in fission, per neutron absorbed in the fuel. It is a property of the fuel alone;
- ϵ is the 'fast fission factor' which lies in the range $1.02 \leq \epsilon \leq 1.08$ depending on the reactor design. It accounts for the extra fission caused mainly by fast neutrons causing fission in non-fissile material such as ${}^{238}_{92}$U in the reactor core;
- $\eta\epsilon(1 - l_f)$ is the number of fast neutrons per neutron absorbed which survive the leakage of fast neutrons through the walls of a finite sized reactor. (The factor $(1 - l_s)$ applies for slow neutron losses.) For a fast reactor neutron reflectors are used to reduce this loss and indeed to control the reactor;
- p is the 'resonance escape probability' for fast neutrons to survive resonance capture in fertile nuclei as they slow to thermal energies. To maintain criticality p should be as close to unity as possible but the $(1-p)$ resonant capture probability results in the *conversion* of a fertile nucleus to a fissile nucleus (see below);
- The neutron utilization factor f is the probability that the thermal neutron is absorbed in the fuel rather than non fissile nuclei in the reactor structure, coolant or moderator. That is $f = \frac{\text{absorb fuel}}{\text{absorb fuel+absorb nonfuel}}$ which again is dependent on the reactor design. This factor will contain the effect of the control rods whose adjustment will make the neutron multiplication factor k close, but below, unity so that reactivity is controlled by the delayed neutrons.

6.1 Conversion Ratio

The conversion ratio C is defined as the ratio of the fissile nuclei produced in fertile nuclei to the the fissile nuclei consumed and is the sum of two components C_{ra} arising from resonance absorption and C_{rth} arising from the thermal neutron flux. We need to calculate how many fissile nuclei are produced from the fertile nuclei for every fissile nucleus destroyed by fission and capture. We do this for the infinite reactor so as to neglect neutron losses through the reactor walls.

Following the absorption of a neutron on a fissile nucleus $\eta_{3,5,9}\epsilon$ fast neutrons are produced and enter the resonance absorption region and there is a probability $(1-p)$ that absorption will occur converting ${}^{238}_{92}$U or ${}^{232}_{90}$Th to fissile nuclei. Thus :

$$C_{ra} = \eta_{3,5,9}\epsilon(1 - p).$$

After the absorption of a neutron on a fissile nucleus, followed by the resonance absorption and slowing down processes we have $\eta_{3,5,9}\epsilon fp$ thermal neutrons per fissile nucleus destroyed. Of these the fraction

$\eta/\eta_{3,5,9}$ are captured on other fissile nuclei. Thus the fraction captured on the fertile nuclei is $(1-\eta/\eta_{3,5,9})$ and so C_{rth} is given by :

$$C_{rth} = \eta_{3,5,9}\epsilon f p\left(1-\frac{\eta}{\eta_{3,5,9}}\right) = \eta\epsilon f p\left(\frac{\eta_{3,5,9}}{\eta}-1\right) = \frac{\eta_{3,5,9}}{\eta}-1 \text{ since } \epsilon\eta f p = 1 \quad \text{for criticality,}$$

and the overall conversion ratio C is given by :

$$C = \eta_{3,5,9}\epsilon(1-p) + \left(\frac{\eta_{3,5,9}}{\eta}-1\right).$$

This is the initial conversion ratio, reflecting the number of fissile nuclei converted from ${}^{238}_{92}$U or ${}^{232}_{90}$Th to ${}^{239}_{94}$Pu or ${}^{233}_{92}$U from the driver fissile material which might be ${}^{233}_{92}$U, ${}^{235}_{92}$U or ${}^{239}_{94}$Pu. If C turns out to be greater than 1 then *breeding* is taking place (see below).

In the EPR reactor (see below) the thermal power is $4.5\ \mathrm{GW_t}$ and the fuel is 100 tonnes of uranium enriched to 5% in the form of 113.5 tonnes of UO_2 (density $11\ \mathrm{tonne/m^{-3}}$). The reactor becomes subcritical when the fissile content falls to 1.5%. $\eta_5 = 2.05$ and for this fuel $\eta_{\text{fuel}} = 1.9$ using the formula in section 4 above. Furthermore in the EPR $p = 0.88$ and $\epsilon = 1.05$, whereby the initial conversion ratio is calculated to be $C = 0.33$.

When N fissile nuclei are consumed CN fissile nuclei are produced from the fertile nuclei and these in turn will produce C^2N fissile nuclei and so on. The total conversion yield is therefore $C + C^2 + C^3 ..... = C/(1-C)$ and the factor for each original fissile nucleus is $1 + C/(1-C) = 1/(1-C)$. So the **fuel utilisation** in the EPR is :

$$y\% = \frac{5\% - 1.5\%}{1-C} = \frac{3.5\%}{1-0.42} = 6.03\% .$$

Note that the *operational* value of the conversion ratio of 0.42 is higher than the initial value, since as the fuel depletes C increases. Thus of the 100 tonnes uranium about 6030 kg of the fuel is consumed and converted into fission fragments leaving 1500 kg fissile nuclei unconsumed.

Over the years engineering advances have been directed at making the lifetime of reactors, albeit under increased neutron fluxes, longer, 60 yrs being the current aim, and to make the dwell time of fuel elements in the reactor as long as possible and to match the reactivity cycle. This is advantageous in consuming and producing through conversion as many fissile nuclei as possible before the reactivity falls too low for criticality, it minimizes the proliferation risk since the growth of ${}^{240}_{94}$Pu makes the fuel unsuitable for bomb making and minimizes the need to re-fuel the reactor. Light water PWR and BWR reactors (see below) have to be shutdown to refuel whereas CANDU and gas cooled reactors can be fuelled in operation. The current aim is for a **fuel burnup** of 60 GWd per tonne of heavy metal fuel in LWR reactors. What does this mean for a 100 tonne uranium (113.5 tonnes uranium dioxide) reactor whose fuel is enriched to 5%? So at a power of $1\ \mathrm{GW_t}$ and a fuel load of 100 tonnes heavy metal, 6000 kg of fissile nuclei must be consumed using the 'ball park' estimate that 1 kg of fissile material must be consumed per day to produce $1\ \mathrm{GW_t}$ for a day. We have seen above that the EPR containing 100 tonnes of uranium enriched to 5% consumes 6030 kg of fissile nuclei before the reactor becomes sub-critical. This is close to the design burnup of the fuel and at the end of the burnup we will have 6030 kg of highly radioactive fission fragments and about 1500 kg of fissile nuclei which could be reprocessed for make new fuel.

If no reprocessing is done to remove/save the plutonium and uranium content then the whole fuel load of 113.5 tonnes is High Level Waste (HLW). With reprocessing the HLW is 6030 kg of very hot fission fragments plus the recuperated fissile plutonium and uranium (1500 kg) and 92.47 tonnes of depleted uranium. At $4.5\ \mathrm{GW_t}$ we must consume 4.5 kg per day of fissile nuclei so one burnup cycle consuming 6000 kg of fissile nuclei lasts for 1333 days or 3.6 yrs. So in a 60 year lifetime there are nearly 17 fueling cycles. If there is no reproceesing then all the 17×100 tonnes (113.5 tonnes UO_2) of the original fuel is HLW. With reprocessing the 17×6.03 tonnes of fission fragments constitute the HLW together with 17×1.5 tonnes of recoverable fissile material.

6.2 Neutron Flux

The neutron flux Φ is the crucial parameter in reactor operation and its connection to d^2N_f, the number of fissions per unit time per unit volume in a reactor is given by :

$$d^2N_f = n_b v A dt \times \frac{\sigma_f}{A} \times n_{tf} A dx = n_b v \sigma_f n_{tf} dV dt \quad \text{, or}$$

$$d^2N_f = n_b v \sigma_f n_{tf} dt dV = R_f dt dV = \Phi \sigma_f n_{tf} dt dV$$

where dV is the core volume element - a result we have previously derived for a general process with cross section σ_i . In the above σ_f is the fission cross section and n_{tf} is the number density of fissile nuclei in the core. We now determine the flux of a reactor containing 100 tonnes of uranium (in the form of 113.5 tonnes of UO_2) enriched to 5% and running at a power of 1 GW_t. We first integrate each side of the above over the volume of the reactor core :

$$\int d^2N_f = dN_f = \int \Phi \sigma_f n_{tf} dt dV \text{ or } \frac{dN_f}{dt} = \Phi \sigma_f n_{tf} V = \Phi \sigma_f N_{tf}$$

where dN_f/dt is the fission rate in the core and N_{tf} is the total number of fissile nuclei in the core, assumed to be in a constant neutron flux.

For the left hand side of the above we know that the fission of 1.08 kg of fissile material (in this case $^{235}_{92}$U will produce an energy of 1 $GW_t d$ or a power output of 1 GW_t for one day, that is $60 \times 60 \times 24 = 8.64 \times 10^4$ seconds and on the right we have 5000 kg of fissile nuclei since the 100 tonnes of fuel is enriched to 5%. Therefore :

$$\frac{1.08}{0.235} \times \frac{6 \times 10^{23}}{8.64 \times 10^4} = \Phi \times 507 \times 10^{-24} \times \frac{5000}{0.235} \times 6 \times 10^{23}$$

whereby $\Phi = 0.493 \times 10^{13}$ cm^{-2} s^{-1}. At a full power of 4.5 GW_t the flux will be just over 2.2×10^{13} cm^{-2} s^{-1} and reactor components, including the pressure vessel, have to be designed to cope with the radiation damage caused by such neutron fluxes over the lifetime of the reactor.

Since η is so large for *fast fission* in the fissile nuclei (see Table 3 above) it is possible to achieve criticality in reactors whose cores contain fissile, or greatly fissile enriched material, using the delayed neutrons to achieve criticality. Note however that criticality in $^{233}_{92}$U and $^{239}_{94}$Pu fuelled reactors is more delicate compared to a $^{235}_{92}$U fuelled reactor since the fraction of delayed neutrons is considerably lower, 0.21% and 0.26% compared with 0.67%. In such *fast reactors* the power density is much higher since fissions occur in the much smaller volume of fuel and requires cooling by liquid metals such as sodium, lead/bismuth or lead. For fast reactors $f \approx 0.7 - 0.8$ where the fast neutrons are lost to the coolant and the support structure.

It is not possible to achieve criticality in a lump of natural uranium because the product $\eta \times \epsilon$ is too small. Also uranium does not effectively slow down neutrons, and as they slow down the neutrons encounter the large resonance cross sections on $^{238}_{92}$U which remove neutrons (see above). However if the natural uranium is distributed homogeneously or heterogeneously in rods, within a *moderator* then criticality can be achieved. The role of the moderator is to slow down the fast neutrons to thermal energies ($KE_t \approx 0.025$ eV) so that they encounter the very large fission cross section and not the large resonance capture cross sections in the fuel.

6.3 The Moderator

The requirements for a moderator are :

- The ability to slow down fast neutrons in as short a distance as possible. Since this influences the size of a reactor the slowing down length is an important parameter determining its capital cost. Neutrons

scatter off nuclei in the moderator so the greater the scattering cross section σ_{ms} of the moderator nuclei and the lighter these nuclei are, the better is the moderator. Neutrons scattering off heavy nuclei rebound with little energy loss. The slowing down property of a moderator is characterised by the number of collisions to slow a $2\ \mathrm{MeV}$ neutron to thermal energies, n_{ms}, which we quote without calculation (see Table 4), and by the scattering length $\lambda_{ms} = \frac{1}{\Sigma_{ms}} = \frac{1}{n_m \sigma_{ms}}$ where n_m is the number density of moderator nuclei. To the extent the slowing down process is a random walk then the mean distance between the origin of the fast neutron and the point it thermalizes, L_s is given by $L_s = \sqrt{n_{ms}}\lambda_{ms}$.

- The moderator should not absorb neutrons as they slow down nor after the neutrons thermalize. Absorption in the moderator is characterised by the absorption length $\lambda_{ma} = \frac{1}{n_m \sigma_{ma}}$ where σ_{ma} is the absorption cross section of the moderator nuclei. Once the neutrons thermalize they scatter off moderator nuclei with a mean path between scatters given by λ_{ms} and in an infinite homogeneous moderator this random walk motion would continue until they are absorbed by moderator nuclei, a process characterised by the absorption length λ_{ma}. Provided λ_{ma} is much greater than λ_{ms} then the number of scatters (n_{mr}) in this random walk is given by $\frac{\lambda_{ma}}{\lambda_{ms}}$ and the distance the neutron is absorbed from its origin as a thermal neutron is given by $\sqrt{n_{mr}}\lambda_{ms}$. This distance is known as the diffusion length L_d and the migration length $L_{mig} = \sqrt{L_s^2 + L_d^2}$ characterises the distance a neutron travels from its origin in fission to the point it is absorbed. It is an important parameter in reactor design since a short migration length implies that fuel elements must be close together to enable criticality. Table 4 summarises the properties of three moderating materials and note (*) that the migration length L_{mig} for water in practice is taken to be $6.25\ \mathrm{cm}$ rather than the $5\ \mathrm{cm}$ calculated in the table because the slowing down process for water is more forward directed than in a random walk process.

- It should be safe under normal and abnormal reactor conditions.

- It is an advantage if the moderator can also serve as the coolant.

- Preferably it should be inexpensive.

Water, heavy water and graphite have been used as moderating materials and are compromise solutions to the above criteria (see Table 4) :

	$\rho\ (\mathrm{g\,cm^{-3}})$	λ_{ms} (cm)	λ_{ma} (cm)	n_{ms}	n_{mr}	L_{mig} (cm)
H_2O	1.00	0.43	51.8	16	120	5*
D_2O	1.10	2.40	13400.0	29	5600	180
C	1.67	2.70	705.0	115	260	52

Table 4: Summary of moderator properties.

7 Reactor Design and Operation

The fuel composition, coolant properties and in thermal reactors, the moderator are all interrelated factors that influence the design and operation of a nuclear reactor.

For **thermal reactors** the coolant determines the rate that heat can be removed from the reactor core and the maximum thermal efficiency. Water cooling can transfer the maximum amount of heat in thermal reactors but the need to keep below the critical point limits the thermal efficiency. (There are proposals to cool reactors with steam well beyond the critical point but many challenges remain to be addressed.) Gas cooling cannot transfer as much heat so the reactor core has to be larger and hence higher capital costs, but higher temperatures and thus higher thermal efficiencies are possible.

The moderator through the migration length L_{mig} determines how close fuel elements have to be to ensure criticality. In (light) water cooled reactors where the coolant is also the moderator the fuel elements within a fuel assembly have to be close packed since the migration length is only 6.25 cm and neutron losses are increased by absorption in the fuel and its cladding. Thus neutron economy is not good and enriched fuel has to be used to ensure criticality. For heavy water the migration length is much larger at 180 cm. and neutron economy is so good that natural uranium can be used as fuel.

Graphite is the moderator most commonly used in gas cooled reactors where the migration length is 52 cm. Neutron economy is very good and natural uranium can be used as fuel. The use of enriched fuel however allows an increase of the power density (provided of course the gas flow and/or the outlet temperature can be increased), leading to a reduced core size and capital cost for the reactor.

There are advantages in using enriched fuel even when the neutron economy is good. Firstly enrichment gives a greater range over which criticality can be established so that more of the fissile nuclei are consumed up to the point where depletion becomes too great to maintain criticality. Secondly the volume of the High Level Waste (HLW) is reduced. The HLW comes from the transuranic nuclei made in capture processes and from the radioactive decay chains of the fission fragments. The amount of this waste is quite similar to the amount of fissile nuclei consumed in reactor operation but unless the fuel is reprocessed to remove uranium and plutonium then all the reactor fuel is HLW. Nowadays enrichment is effected by cascades of centrifuges.

It is very instructive to consider the start up of a reactor and then to follow the operation since the conditions of criticality change during the fuel burnup.

Once the reactor is fuelled the source of the initial neutrons is the spontaneous fission in $^{238}_{92}\mathrm{U}$ or a neutron source. The initial neutron flux is very low and so the fission rate and the delayed neutron rate are also very low. The control rods have to be carefully withdrawn allowing the power and the neutron flux to slowly rise and for the reactor to remain critical on the delayed neutrons. Potentially this is the most dangerous phase of reactor operation since any positive reactivity excursion arising from a statistical fluctuation in the flux could be greater than the delayed neutrons and require the rapid re insertion of the control rods. Once the design power level is reached the control rods then are adjusted to short and longer term reactivity changes. In PWR reactors much of the reactivity control is effected by Boric acid (boron is a neutron absorber) dissolved in the coolant water.

Concerning short term reactivity changes it is important that the overall rate of change of reactivity be negative with respect to temperature. Higher temperatures cause the fuel, coolant and moderator densities and consequently absorption processes, to decrease which in turn would appear to increase the reactivity. However in water cooled reactors the decrease in density also results in the more important decrease of moderation which leads to a decrease in reactivity. Higher temperatures also increase the vibrations of nu-

clei in the fuel which causes the Doppler broadening of narrow resonant cross section in the fertile nuclei leading to increased absorption and thus a reduced reactivity. Indeed in light water cooled reactors should there be a complete loss of the coolant the reactor would become sub critical and stop because the is no moderation. **However it is vital to cool the core because the decay of the fission fragments continue to produce heat.** Positive temperature reactivity change occur when for example fuel is cooled by water and the neutrons moderated by graphite as in the RBMK reactors in the former Soviet Union. Should the coolant produce steam voids then the absorption of neutrons decreases and more neutrons pass into the moderator where they thermalise and increase the reactivity. There is also a positive void coefficient in the heavy water CANDU reactor which we shall discuss later.

Longer term reactivity changes are caused by the depletion of the fissile content of the fuel, the build up of reactor poisons from the decay of the fission fragments and the growth of fissile content of the fuel caused by conversion of ${}^{238}_{92}$U (or ${}^{232}_{90}$Th) into ${}^{239}_{94}$Pu (or ${}^{233}_{92}$U). In steady state operation the neutron flux both creates the reactor poisons and consumes them so that an equilibrium concentration is set up. The growth of the reactor poison ${}^{135}_{54}$Xe after start up will reduce the reactivity of a reactor by about 3% and when a reactor is shut down its concentration at first grows because it is not being destroyed by the neutron flux. The reactivity falls to -15% before the ${}^{135}_{54}$Xe decay, with a $9\ \mathrm{hr}$ decay lifetime, dominates. So when a reactor is shut down it's not possible to start up again for about a day until the ${}^{135}_{54}$Xe has decayed away. The neutron flux will also cause fissions in the ${}^{239}_{94}$Pu (or ${}^{233}_{92}$U) which contributes some 35% in LWRs to the power output depending on the 'burnup' (as we have seen above) and also convert the ${}^{239}_{94}$Pu to heavier isotopes which is important regarding proliferation issues. When ${}^{233}_{92}$U is made from ${}^{232}_{90}$Th it is accompanied by the production of ${}^{232}_{92}$U which is highly radioactive so both reactor produced plutonium (provided the dwell time is long enough) and ${}^{233}_{92}$U are not easily separated for bomb making which is important from the point of view of nuclear proliferation. No doubt special steps could circumvent this difficulty.

In a **fast reactor** there is no moderator and the fuel in the core is 20% ${}^{239}_{94}$Pu and 80% ${}^{238}_{92}$U with the aim of exploiting the large η value of ${}^{239}_{94}$Pu (2.9) to **breed** more ${}^{239}_{94}$Pu in a ${}^{238}_{92}$U blanket that surrounds the core, than is consumed in the reactor. With an η of 2.4 ${}^{233}_{92}$U would be a better fast reactor fuel than ${}^{235}_{92}$U ($\eta = 2.35$) which has been used as a fuel in an experimental fast breeder reactor. The core is extremely compact and heat has to be removed by liquid metal coolants like sodium (as in shut down European fast reactors), lead/bismuth or lead as in some of Russia's nuclear submarines. Because there are only 0.26% delayed neutrons in plutonium fission the reactor is designed to operate just critical on the delayed neutrons. An upward temperature excursion would cause reactor components to expand leading to a reduction in the fissile number density and to the components moving apart allowing greater neutron loss, both effects leading to subcriticality. The Doppler broadening of neutron absorbing cross sections in the core's ${}^{238}_{92}$U also has a negative effect on reactivity. (This is the reason why the ${}^{239}_{94}$Pu is not allowed to exceed 30% of the core mass.) Criticality conditions are set by neutron reflectors surrounding the reactor and control rods inserted into the core all of which can be used in emergency conditions.

The **breeding ratio** B, the rate of fissile material production in the blanket, divided by the rate that fissile material is consumed in the core, has to satisfy the condition $B < (\eta - 1)$ since there must always be one neutron to initiate the next fission and to keep the chain reaction going and B has to be greater than 1 to create more fuel than is consumed. In practice η has to be greater than 2.2 to allow for neutron losses so that breeding can take place. If m is the amount of fissile material consumed in a day at $1\ \mathrm{GW_t}$ and P_0 is the thermal power in $\mathrm{GW_t}$ then the **breeding gain** is given by $(B-1)mP_0\ \mathrm{kg\ d^{-1}}$. For **fast reactors** the η values and the fissile consumption amounts are obtained from the cross sections in the fast neutron Table 3. The inelastic cross sections σ_{in} do **not** contribute since there is no loss of neutrons, the neutrons lose energy and a softer neutron energy spectrum enters the breeding blanket. The consumption numbers for **fast reactors** are $1.14\ \mathrm{kg}$ for ${}^{233}_{92}$U, $1.16\ \mathrm{kg}$ for ${}^{235}_{92}$U and $1.11\ \mathrm{kg}$ for ${}^{239}_{94}$Pu per day to produce a power of $1\ \mathrm{GW_t}$ for a day or an energy of $1\ \mathrm{GW_t d}$.

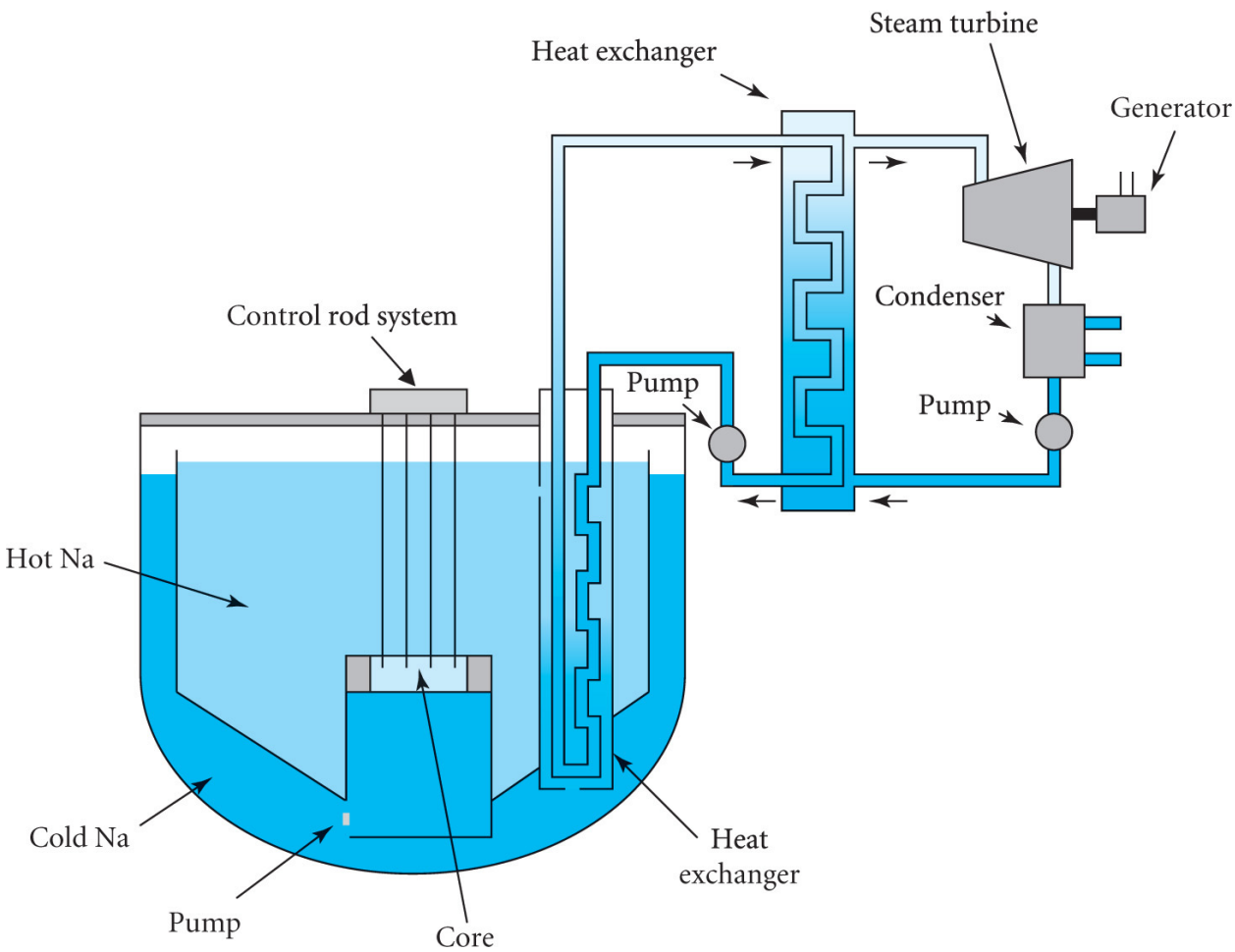

Figure 1: A pool type fast breeder reactor

Suppose we have a fast breeder operating at a power of $1\ \mathrm{GW_t}$ and containing in its core 1000 kg of ${}^{239}_{94}\mathrm{Pu}$ and 4000 kg of ${}^{238}_{92}\mathrm{U}$ surrounded by a thick blanket of ${}^{238}_{92}\mathrm{U}$. By the time the neutrons reach the blanket, the neutron flux is considerably lower than the core and the neutrons have lost sufficient energy so that they do not cause fast fission in ${}^{238}_{92}\mathrm{U}$ and have a very high probability of capture in ${}^{238}_{92}\mathrm{U}$ and being converted to ${}^{239}_{94}\mathrm{Pu}$. If the ${}^{239}_{94}\mathrm{Pu}$ remains in the blanket some will be transformed into heavier plutonium isotopes and the doubling time in days will be the initial charge of fissile fuel (1000 kg) divided by $(B-1)mP_0$ where P_0 is in this case $1\ \mathrm{GW_t}$, m is 1.11 kg per day and $(B-1)$ is 0.1 according to the design of the reactor. The doubling time is close to 25.5 yrs. However leaving the bred plutonium in the blanket is not its best use. In principle it could be immediately used to fuel another reactor and if the bred plutonium were to be regularly removed from the blanket to fuel other reactors then the doubling time would be considerably reduced. This plutonium would be weapon grade.

Breeding is important since at current rates the supply of uranium may last $80-100$ yrs where we are using only the 0.7% fissile component plus some more from the conversion process. In principle breeding would allow the use of all the uranium but note the long doubling time. Breeding ${}^{233}_{92}\mathrm{U}$ from thorium, which is $3-5$ times more abundant than uranium in a CANDU or molten salt thermal reactors (see below) is the other and maybe more practical, long term goal.

8 Nuclear Assets ?

Following the weapons programme of the cold war it is estimated that the world stock of **military** plutonium is 250 tonnes and that of highly enriched uranium 1750 tonnes. In Russia and the USA some of this stockpile is being diverted to the civil nuclear programme and clearly its total disposal to civil use would be the ultimate 'beating of swords into ploughshares and spears into pruning forks'.

In part due to the British decision to follow the plutonium route to nuclear weapons from which emerged the MAGNOX generation of graphite moderated, CO_2 cooled reactors and in part due to the failure of the THORP reprocessing plant in Sellafield, the UK has about a 100 tonne stock of plutonium containing 68% $^{239}_{94}$Pu. This represents about two thirds of the world civil stock but its content of $^{239}_{94}$Pu at 68% is considerably higher than the 53% average of stock outside the UK. Although weapon grade plutonium requires a 93% purity of $^{239}_{94}$Pu it is widely accepted that civil plutonium could be used to make a nuclear weapon, albeit a large one with a relatively small and unreliable explosive yield.

9 High Level Nuclear Waste

Before discussing specific reactors we now discuss the High Level Nuclear Wastes produced by reactors. After a 60 GWd per tonne burnup in the EPR reactor we have seen that 100 tonnes of the heavy metal fuel (113.5 tonnes of uranium dioxide) enriched to 5% will have fissioned 6030 kg of fissile nuclei so that 6030 kg of highly radioactive fission fragments, and 1500 kg of fissile nuclei remain in addition to the 92.47 tonnes of $^{238}_{92}$U. At a power of 4.5 GW_t a burnup cycle corresponds to about 1333.3 d or 3.65 yr of reactor operation. At the end of each cycle, if no reprocessing takes place then the reactor fuel 100 tonnes of heavy metal, 113.5 tonnes of uranium dioxide, constitutes the High Level Nuclear Waste. With recycling the 6030 kg fission fragments plus any fissile material and long lived actinides left after reprocessing would constitute the HLW. Reprocessing would recover most of the remaining 1500kg of fissile material. The spent fuel rods must be stored in cooled ponds for one to three years before they can be moved for reprocessing yielding vitrified waste, or further storage. The volumes of High Level Waste are not large and are small if reprocessing takes place since the uranium oxide and plutonium are separated from the long lived actinides and the really hot 6030 kg of fission fragments ; environmental concerns have made long term disposal problematic but it seems to me to be a soluble problem. Indeed the French will by 2025 commision the Meuse/Haute-Marne repository 500 m deep in mudstone at Bure where vitrified waste after a cooling period of 50yrs. will be stored long term.

Regarding **reprocessing** to create mixed oxide ie uranium and plutonium oxides (MOX) fuel, it should be pointed out that France has succeeded in setting up an industrial scale plant at La Hague on the Cotentin peninsula in Normandy whereas the THORP plant at Sellafield is the largest industrial disaster in the UK costing £1.3billion. In 2005 the THORP plant suffered a major leak, fortunately contained, of nitric acid containing 20 tonnes of uranium and 160 kg of plutonium. It was expected that the plant would reprocess 560 tonnes of nuclear fuel but during the nine years of operation only 15 tonnes of reprocessing was achieved.

Recently the UK government have cancelled the contract for an international team of contractors to clean up Sellafield because of cost overruns and poor performance and to take the project 'in house'.

10 Light Water Reactors - the PWR and BWR

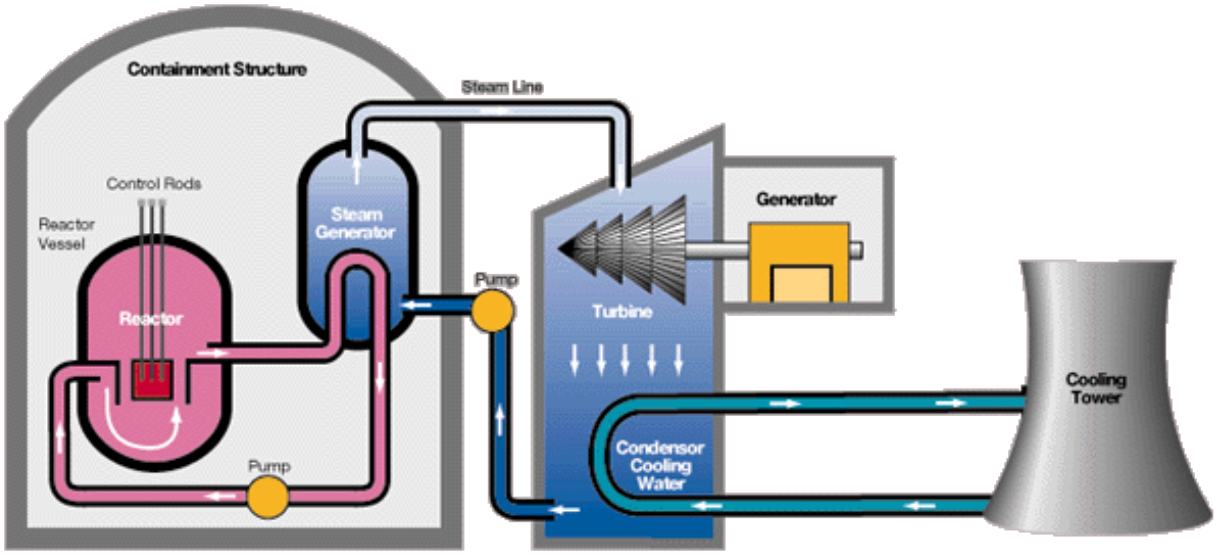

Figure 2: The PWR reactor

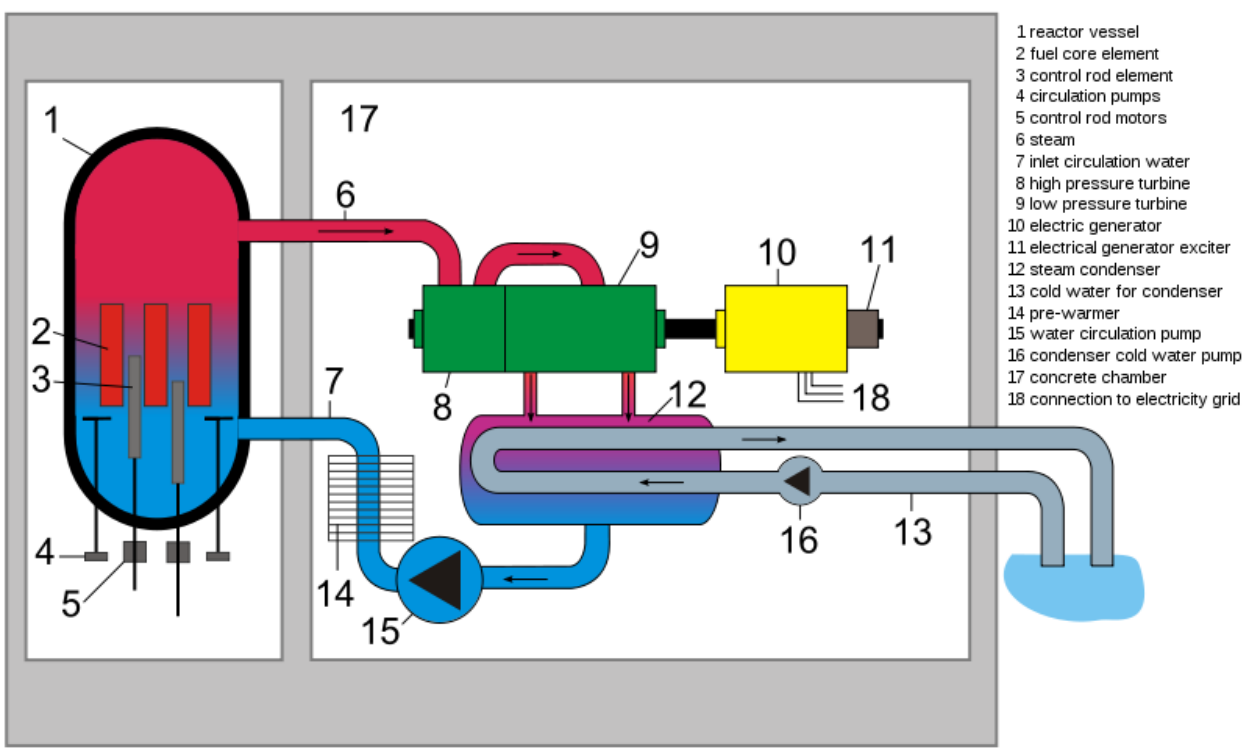

Figure 3: The BWR reactor

The simplified diagrams of the PWR and BWR reactors show clearly the main difference between the two reactor types

- in the BWR the primary coolant circuit water is allowed to boil and used directly to raise steam for the turbine. The pressure in the primary circuit is roughly half that in the PWR and thus the pressure vessel construction is less costly. This cost saving is partly offset because the pressure vessel in a BWR has to be larger than that of a PWR since the energy density in a BWR is lower, reflecting the lower heat transfer of the BWR.

- The BWR avoids the costs of the complicated secondary heat exchanger circuit.

- All pipes etc must be shielded in the turbine hall of a BWR since failure in the fuel elements results in radioactivity getting directly into the steam for the turbine.

- the control rods in the BWR have to be inserted upwards into the core because control is not so effective in the upper water/steam region, so that in emergency control rods cannot be dropped into the core under gravity as in the PWR.

Following shutdown it is essential in both reactor types to cool the core because the fission fragments produce heat (some 6.5% at shutdown) from their radiocative decay and for the same reason to cool the spent fuel for some one to three years before the spent fuel rods can be sent to further storage or reprocessing. These requirements together with the dangers of the steam-metal explosion have been previously emphasized in these lectures.

10.1 The EPR Reactor

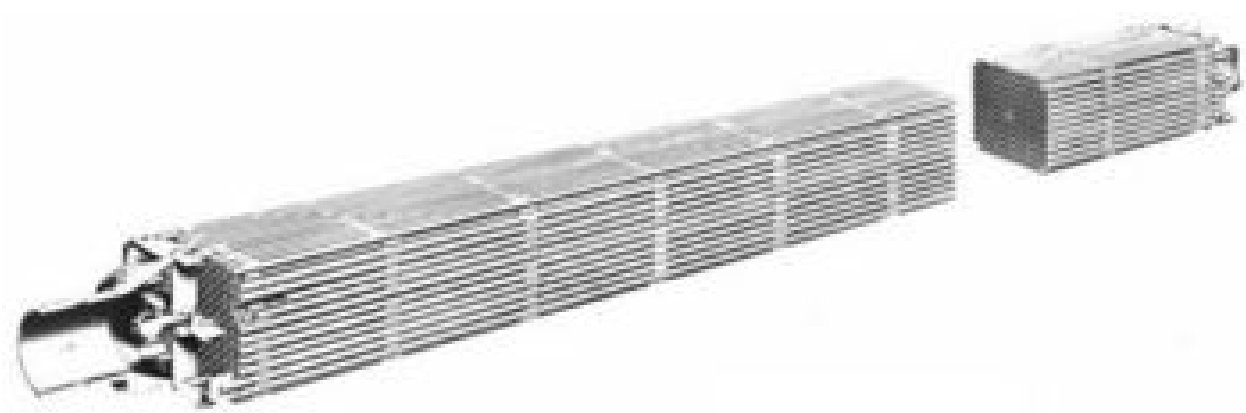

Figure 4: The EPR fuel assembly

The EPR (the European Pressurized Reactor or the Evolutionary Pressurized Reactor) reactor is the current stage of development of the PWR reactor. Four such reactors are under construction one in Flamanville, France, an other at Olkiluoto, Finland, two in Taishan, China and it is proposed to build two or more in the UK. There have been major constuction delays and cost overruns at Flamanville, Olkiluoto and China. To cap it all, serious deficiences have been reported in the steel of parts of the Flammanville pressure vessel.

It has the following features -

Fuel - 100 tonnes of Uranium enriched to 5% in the form of 113.5 tonnes of UO_2 (density $11\ \mathrm{tonne\ m^{-3}}$) As we have seen before $\eta = 1.9$ and C = 0.33.

Core - Right cylinder, height $4m$ radius $1.85m$.

Cooling - Water at 155 bar and $320°C$ where keeping below the critical point limits the thermodynamic efficiency.

Steam generation - Heat is transferred from the primary cooling circuit through the heat exchanger into a secondary steam generating circuit (see below).

Moderator - Water, same as the cooling. Note how the short migration migration length makes it necessary for the fuel elements to be close together in the fuel assembly as shown in the EPR fuel assembly.

Control - is effected by neutron absorbing rods and boric acid dissolved in the coolant.
Thermal power - $4.5\ \mathrm{GW_t}$.
Electrical power - $1.6\ \mathrm{GW_e}$.
No. of fuel assemblies - 241.
No. of primary steam loops - 4. There are four primary coolant circuits each feeding secondary steam generating circuits which supply the turbines with steam at $280°C$. This minimizes radiation in the turbine hall.
Thermodynamic Efficiency - Actual - 36%, Carnot - 45%.
Burnup - $60\ \mathrm{GWd\ tonne^{-1}}$.
Service Life - 60 years.

11 The CANDU Reactor

The Calandria

This reactor **avoids the use of a large pressure vessel** since the fuel is held in a calandria of high pressure horizontally mounted tubes containing the coolant and the fuel elements.
The arrangement allows **refueling on line** ie when the reactor is operating.

The moderator and control rods

The moderator is heavy water at atmospheric pressure which is **expensive** but has **excellent neutron economy**, allowing the possibility of **conversion** using the Thorium cycle.
Tritium is produced in the heavy water which has to be removed to avoid radiation exposure.

The Coolant Circuit and the Fuel Elements

The coolant is heavy water which at $310°C$ and 12 bar passes through heat exchangers producing steam at $266°C$ - the CANDU - 6 reactor has a thermal output of $2064\ \mathrm{GW_t}$ and an electrical output of $745\ \mathrm{GW_e}$ ie a thermodynamic efficiency of 36% compared with a Carnot efficiency of 45.5%

Since the coolant heavy water is separate from the heavy water in the moderator the CANDU Reactor has a **positive void coefficient**. Fast neutrons emerging from the fuel can be slowed down in the coolant into the resonance region and there is a small loss of neutrons here. However small this loss, if there are voids in the coolant this loss is reduced and the reactivity increases. Reactor design guards against this positive void coefficient.

Since the CANDU series of reactors use natural uranium the **volume of high level nuclear waste for a given power output** is greater than for reactors using enriched fuels. The η and the conversion ratio for natural uranium has already been worked out above.

The $^{235}_{92}$U/$^{233}_{92}$U/$^{232}_{90}$Th cycle
The excellent neutron economy makes possible the use of $^{235}_{92}$U as the driver fuel to convert some of the abundant fertile $^{232}_{90}$Th nuclei into $^{233}_{92}$U which can then be used as the fissile nuclei. Assuming we use 2% $^{235}_{92}$U as the driver fuel and 98% $^{232}_{90}$Th as the fertile nuclei then we find $\eta = 1.33$ and $C = 0.85$ with $p = 0.85$ and $\epsilon = 1.0$. Clearly this is a high conversion factor but actual performance must depend on a

more sophisticated analysis than is done here.

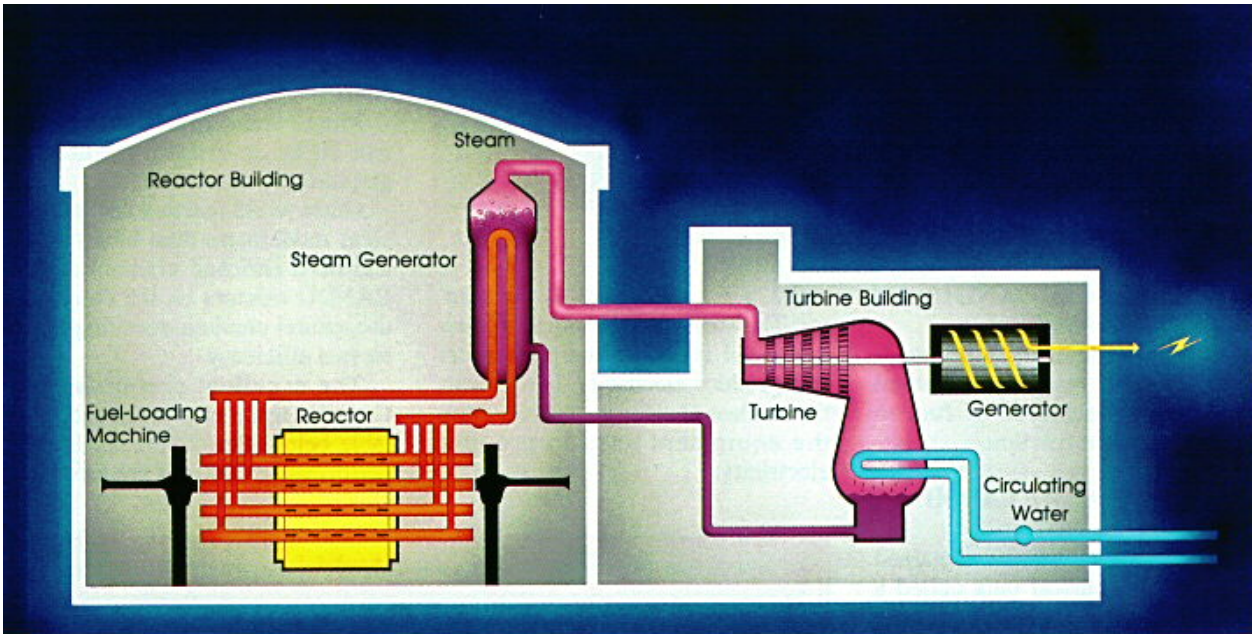

Figure 5: The CANDU reactor

12 Thorium Fuelled Molten Salt Reactors

There have been recent reports that China is pursuing the Thorium Molten Salt Reactor technology pioneered by Alvin Weinberg at the Oak Ridge National Laboratory 1954 - 1974.

The Thorium cycle is based on the following reaction

$$n + {}^{232}_{90}\mathrm{Th} \rightarrow {}^{233}_{90}\mathrm{Th}(22\mathrm{m}) \xrightarrow{\beta} {}^{233}_{91}\mathrm{Pa}(21\mathrm{d}) \xrightarrow{\beta} {}^{233}_{92}\mathrm{U}(1.6 \times 10^5\mathrm{yr})$$

The difficulty associated with the protactinium is that if the 21 day half-life ${}^{233}_{91}$Pa be left in the reactor then much of it is consumed by the neutron flux, thereby reducing the yield of ${}^{233}_{92}$U.

	${}^{233}_{92}$U	${}^{232}_{90}$Th	
σ_f (barns)	81	0	
σ_c (barns)	10	1.6	
ν	2.5	0	
% Delayed neutrons	0.26	0	

Table 5: **Epithermal neutron** spectrum averaged cross sections and related parameters

In such a proposed reactor molten salt (melting temperature about 500°C) containing 1.1 tonnes of ${}^{233}_{92}$U (or another fissile material for startup) and 65 tonnes of ${}^{232}_{90}$Th enters the reactor vessel at 550°C and leaves at 650°C. This primary salt circuit transfers heat to a secondary molten salt circuit which in turn supplies heat to a Rankine Steam Cycle working between 500°C and 30°C. Graphite rods are used to partially moderate the neutrons to epithermal energies where the relevant cross sections and nuclear properties are given in the

table. The salt is only critical in the region defined by the graphite moderating rods.

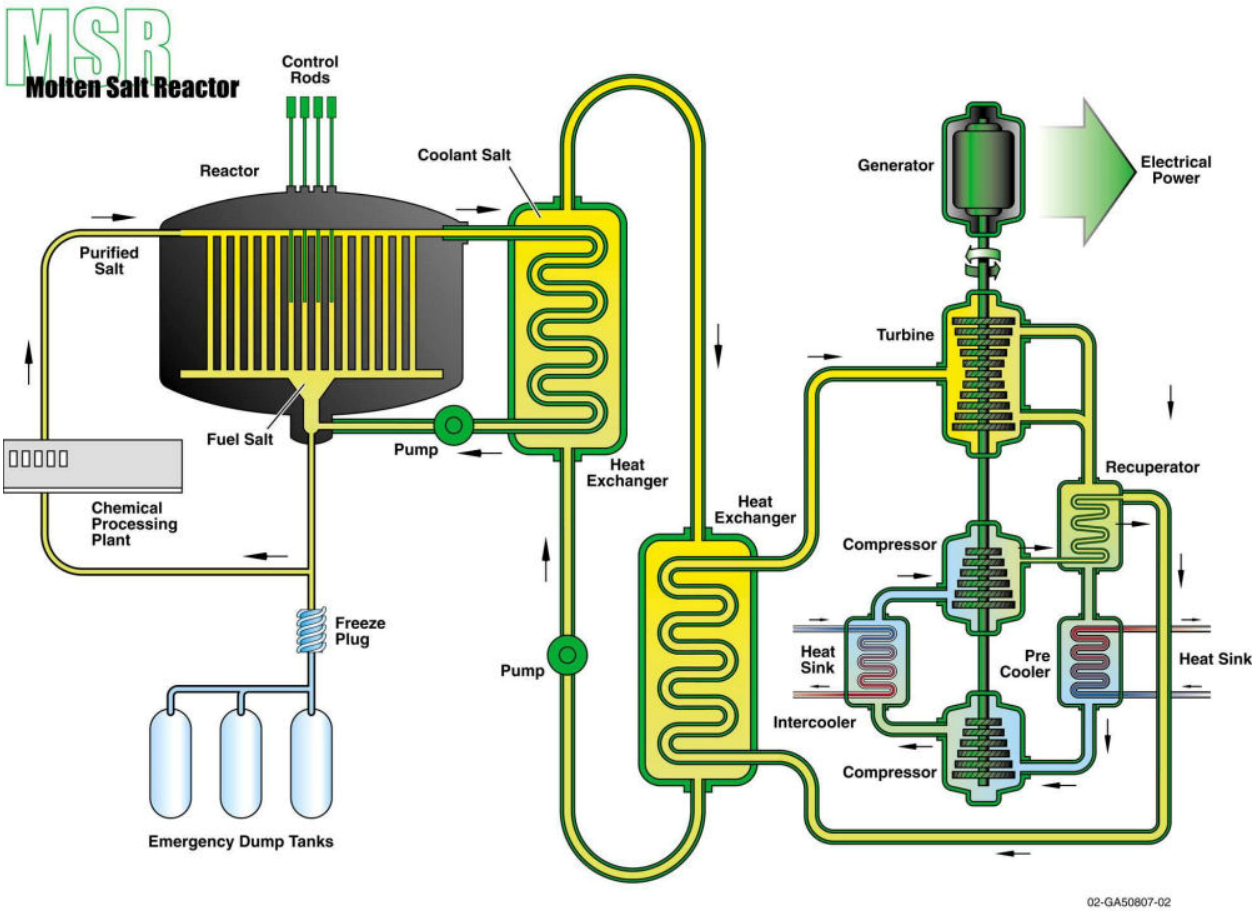

Figure 6: The Molten Salt Thorium Reactor

For the ideal (Carnot) thermal efficiency we have $\epsilon = 1 - 303/773 = 0.61$

For η_3 we have $\eta_3 = 2.5 \times 81/(81 + 10) = 2.2$.

For η_{fuel} we have $\eta_{fuel} = (2.5 \times 81 \times 1.1)/(1.1 \times (81 + 10) + 65 \times 1.6) = 1.09$.

For C with $p = 0$ we have $C = \eta_3/\eta_{fuel} - 1 = 1.03$ indicating that this reactor is just about a breeder.

Since the reactor works at $2.5\ \mathrm{GW_t}$ and we can assume that the fissioning of $1.08\ \mathrm{kg}$ per day of ${}^{233}_{92}\mathrm{U}$ produces $1\ \mathrm{GW_t}$ for a day (or an energy of $1\ \mathrm{GW_t d}$) then we need to fission $2.5 \times 1.08\ \mathrm{kg}$ per day. Thus

$$\frac{2.5 \times 1.09}{0.235} \times \frac{6 \times 10^{23}}{8.64 \times 10^4} = \Phi \times 81 \times 10^{-24} \times 10^{-24} \times \frac{1100}{0.235} \times 6 \times 10^{23}$$

whence $\Phi = 3.5 \times 10^{14}\mathrm{cm}^{-2}\mathrm{s}^{-1}$.

What are the positive aspects of the thorium reactor and its cycle?

Thorium is seven times more abundant than uranium.

It is more difficult to make an A-bomb with ${}^{233}_{92}$U since in the thorium cycle it is normally contaminated with ${}^{232}_{92}$U which is a strong γ emitter, although the critical mass is much smaller than that for ${}^{235}_{92}$U.

The thorium cycle produces far less long lived actinides - thorium is just three nucleons short of ${}^{235}_{92}$U!

The reactor operates at atmospheric pressure thus avoiding the need for a large pressure vessel.

In the event of a containment failure the coolant does not boil with a huge increase of volume, rather it solidifies.

The reactivity is controlled externally by appropriately mixing the fissile and fertile nuclei.

An upward reactivity excursion would lead to the salt temperature increasing, the graphite and salt moderation being reduced, leading to a negative reactivity temperature coefficient.

The relatively high temperature of the reactor improves the thermodynamic efficiency of the steam cycle.

The reactor is a near breeder.

13 The Fukushima Nuclear Accident

It would be impossible to conclude a section on nuclear power without referring to the March 11th. 2011 tragic disaster in Japan, where the nuclear accident followed the devastation caused by the tsunami and earthquake which resulted in 18.000 deaths. Although few if any deaths were attribable to the nuclear disaster some $150,000$ people were evacuated from the prohibited zone around the reactors.

The simplified diagrams of the PWR and BWR reactors show clearly the main difference between the two reactor types - in the BWR the primary coolant circuit is used directly to raise steam for the turbine. Following shutdown it is essential in both reactor types to cool the core because the fission fragments produce heat (some 6.5% at shutdown) from their radiocative decay and for the same reason to cool the spent fuel for some one to three years before the spent fuel rods can be sent to further storage or reprocessing. These requirements together with the dangers of the steam-metal explosion have been previously emphasized in these lectures. I was also unaware that in Japan there are two electricity grids, the East Japan grid working at 50Hz originating from AEG German technology in 1895 and the West Japan grid at 60Hz originating from GE US technology in 1896 which requires interconnection by HVDC to be discussed later in this course in the section on offshore wind generation.

The Fukishima nuclear reactor disaster is fully described in the Wikipedia article from which I have taken the first paragraph and urge you to read the whole article.

The Fukushima Daiichi nuclear disaster , Fukushima Dai-ichi is a series of equipment failures, nuclear meltdowns, and releases of radioactive materials at the Fukushima I Nuclear Power Plant, following the Tooku earthquake and tsunami on 11 March 2011. It is the largest nuclear disaster since Chernobyl in 1986. On 16 December 2011 Japanese authorities declared the plant to be stable, although it would take decades to decontaminate the surrounding areas and to decommission the plant altogether. The plant comprises six

separate boiling water reactors originally designed by General Electric (GE), and maintained by the Tokyo Electric Power Company (TEPCO). At the time of the quake, Reactor 4 had been de-fuelled while 5 and 6 were in cold shutdown for planned maintenance. The remaining reactors shut down automatically after the earthquake, and emergency generators came online to control electronics and coolant systems. The tsunami broke the reactors' connection to the power grid, leading the reactors to begin to overheat. The flooding and earthquake damage hindered external assistance. In the hours and days that followed, reactors 1, 2 and 3 experienced full meltdown. As workers struggled to cool and shut down the reactors, several hydrogen explosions occurred. The government ordered that seawater be used to attempt to cool the reactors - this had the effect of ruining the reactors entirely. As the water levels in the fuel rods pools dropped, they began to overheat. Fears of radioactivity releases led to a 20 km (12 mile)-radius evacuation around the plant, while workers suffered radiation exposure and were temporarily evacuated at various times. Electrical power was slowly restored for some of the reactors, allowing for automated cooling. Japanese officials initially assessed the accident as Level 4 on the International Nuclear Event Scale (INES) despite the views of other international agencies that it should be higher. The level was successively raised to 5 and eventually to 7, the maximum scale value. The Japanese government and TEPCO have been criticized in the foreign press for poor communication with the public and improvised cleanup efforts. On 20 March, the Chief Cabinet Secretary Yukio Edano announced that the plant would be decommissioned once the crisis was over. Japan's foreign minister criticized foreign media reports over the accident as overdone and excessive.

The Japanese government estimates the total amount of radioactivity released into the atmosphere was approximately one-tenth as much as was released during the Chernobyl disaster. Significant amounts of radioactive material have also been released into ground and ocean waters. Measurements taken by the Japanese government 30 - 50 km from the plant showed radioactive caesium levels high enough to cause concern, leading the government to ban the sale of food grown in the area. Tokyo officials temporarily recommended that tap water should not be used to prepare food for infants. There were no deaths or serious injuries due to direct radiation exposures, but a few of the plant's workers were severely injured or killed by the disaster conditions resulting from the earthquake. At least six workers have exceeded lifetime legal limits for radiation and more than 300 have received significant radiation doses. Future cancer deaths due to accumulated radiation exposures in the population living near Fukushima have been estimated to be between 100 and 1,000. Fear of ionizing radiation could have long-term psychological effects on a large portion of the population in the contaminated areas. Decommissioning of the plant will likely take 40 years to complete.

	CCGT	CCGT CCS	ASCOCCS	EPR
Cap. Cost ($\pounds/\mathrm{kW_e}$)	808	1370	3000	5000
Am. Cap. Cost ($\pounds/\mathrm{MWh_e}$)	9	21	37	70
Fuel ($\pounds/\mathrm{MWh_e}$)	49	56	36	5
Carbon ($\pounds/\mathrm{MWh_e}$)	18	4	5	0
CO2 Capture/Storage ($\pounds/\mathrm{MWh_e}$)	0	4	5	0
Decommision/ Waste	0	0	0	2
O+M	4	10	22	14
Elec. Gen. Cost ($\mathrm{p/kWh_e}$)	8	9.5	10.5	9

Table 6: Summary of Fossil Fuel and Nuclear Power Plants (see Energy Generation notes)

There will clearly now be a period of reflection but I would be surprised if these tragic events will lead to a move away from nuclear power even in Japan and other regions prone to natural disasters. In the absence of large rivers Japan has to locate their reactors near the sea for cooling purposes which exposes them to both earthquake and tsunami dangers. It is of interest to note that the inland reactors in France have to be switched off in hot summers otherwise the rivers would become too warm. Germany has announced a retreat from nuclear power which has repercussions in the UK since the German companies E.ON and RWE

Npower who had planned to build Westinghouse - Toshiba reactors (in the absence of UK capability) at Wylfa on the north coast of Anglesey. The Japanese company Hitachi subsequently bought this consortium and was seeking regulatory approval before the suspension of the project in 2019. Even in France, whose nuclear industry has nearly completed the full cycle from fuel preparation, power generation right through to the disposal of nuclear waste, there are now moves to reduce reliance on nuclear power (from 80% down to 60% of supply). The increased demand from Japan for liquified natural gas (LNG) has increased the cost of such imports and energy companies in the UK are reverting to cheap coal from the US following the fracking revolution, to generate electricity in the face of rising gas costs and weak demand.

In the UK the current wholesale price of electricity is around $£45/\mathrm{MWh_e}$ (or $4.5\mathrm{p/kWh_e}$) - below the CCGT price in Table 6, even when the currently unapplicable carbon price is stripped out, since current generation relies on older gas and coal fired stations. EDF (who lost its UK partner Centrica but succeeded in finding Chinese collaborators) have negotiated with HM Government a floor price of $£92.5/\mathrm{MWh_e}$ (or $9.25\mathrm{p/kWh_e}$) for the electricity generated in the thirty five years after the ten years or so construction period of Hinkley Point C. Austria is taking legal action over this agreement claiming unfair state aid and there is disquiet amongst the UK and Chinese authorities concerning delays and cost overruns and the financial viability of the French consortium, let alone the recent news from Flammanville. But HMG was not in a strong position to resist because there is no longer a UK nuclear build capacity and reliance has to put on French, Russian, Korean, Japanese/American and Chinese nuclear constructors. The generation margin of 15% above peak demand is going to fall to 5% in the next three years because coal fired stations which do not meet EU $\mathrm{SO_x}$ and $\mathrm{NO_x}$ emission targets and old nuclear power stations are being closed down. EDF has recently announced that the lifetime of some of the UK's nuclear power stations are going to be extended by ten years. The UK must also make progress in developing a HLW repository.

For the nuclear industry to prosper globally beyond the current largely LWR technology which might last more than a century, there must be a concerted effort to develop the thorium cycle to ensure a long term fuel supply.

14 Small Modular Reactors, Accelerator Driven System and Fusion ?

Given the problems encountered by scaling up large reactor designs there is interest in smaller modular designs that could be industrially manufactured and then installed about the electricity grid, closer to the users. Surely there will safety and environmental objections and for any new nuclear project in particular there is a very long lead time.

Regarding the other two projects since none of these is likely to be power producing in the next 20 years at least we do not discuss them further in this course.